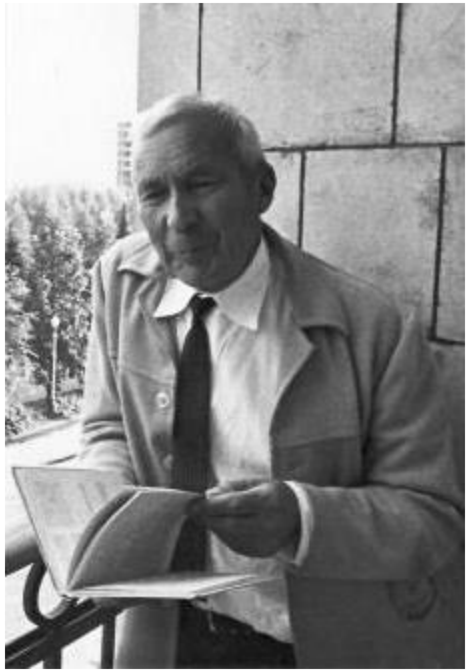
Math Primer
(15. Probability Theory)

Contents



A. Kolmogorov
(1903-1907)

σ −algebra

- The study of probability begins with a ‘random experiment’. For example, consider a random a coin toss experiment. Let’s understand some very important definitions

$\Omega = \{H, T\}$, $\omega \in \Omega$

ϕ, $\{H\}$, $\{T\}$, Ω

true true

Outcome of the experiment ω : $\omega = H$ or T

Sample space Ω (set of all possible outcomes) : $\Omega = \{H, T\}$ $\omega = H$

Event A (any sub-set of Ω) : e.g., $A = \phi$ or $A = \{H\}$

(We say Event A occurred if the outcome belongs to A , i.e., $\omega \in A$)

Similarly, if our experiment consists of two tosses, then

Outcome $\omega = HH$ or HT or TH or TT TT

Sample space $\Omega = \{HH, HT, TH, TT\}$

Event $A = \{HH, HT\}$ ... first toss is head or $A = \{HH, HT, TH\}$ ... at least one head

σ −algebra

PEAKS2TAILS

- Consider a collection $\mathcal{F}$ of sub-sets of Ω (events) which satisfies the following properties

A

I. $\phi \in \mathcal{F}$

II. For every $A \subseteq \Omega$, $A \in \mathcal{F} \Longrightarrow A^C \in \mathcal{F}$

III. For every $A, B \subseteq \Omega$, $A, B \in \mathcal{F} \Longrightarrow A \cup B \in \mathcal{F}$

IV. If $\{A_n : 1,2,3\ldots\}$ is a collection of sub-sets of Ω with $A_n \in \mathcal{F}$ for each $n >= 1$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$

countable infinite

$\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ } – countably infinite

$\mathbb{R}/\mathbb{Q}^c$ – uncountably infinite

$\phi^c = \Omega$

ϕ, Ω

The last one basically says it's closed under countable union. If a collection of sub-sets of Ω satisfies the first three, then it's called an algebra in Ω. If it also satisfies the 4th condition, it's called a σ −algebra. The pair $(\Omega, \mathcal{F})$ is called a measurable space.

Exercise - 1. If $\Omega = \{1,2,3,4\}$ and $\mathcal{F} = \{\{1\},\{2,3\},\{4\},\{2,3,4\},\{1,2,3\},\{1,4\},\phi,\Omega\}$, is $\mathcal{F}$ a σ −algebra on Ω ?

Exercise - 2. Show that if A and B are in the σ −algebra, then $A \cap B$ and $A - B$ are also in the σ −algebra.

σ −algebra

- Consider a random experiment consisting of tossing a coin 3 times. The sample space is given as

8

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Some important σ −algebra are

nothing everything

$\mathcal{F}_0 = \{\phi, \Omega\}$ ... trivial σ-algebra.

A_H A_T

$\mathcal{F}_1 = \{\phi, \Omega, \{HHH, HHT, HTH, HTT\}, \{THH, THT, TTH, TTT\}\}$

$\mathcal{F}_2 = \{\phi, \Omega, \{HHH, HHT\}, \{HTH, HTT\}, \{THH, THT\}, \{TTH, TTT\},$

and all sets that can be built taking unions of these $\}$

$\mathcal{F} = \mathcal{F}_3 = \{$all subsets of $\Omega\}$ ... power set $\mathcal{P}(\Omega)$ $2^8 \to 256$

σ −algebra

- Let's simplify the notations by defining some events

$V_0 = \mathbb{E}^{Q}\left[e^{-rT} V_T \mid \mathcal{F}_0\right]$

$\mathcal{F}_t$

$A_H = \{HHH, HHT, HTH, HTT\}$ … head in the first toss
$A_T = \{THH, THT, TTH, TTT\}$… tail in the first toss

$\boxed{\mathcal{F}_1} = \{\phi, \Omega, A_H, A_T\}$ First toss = H

$\boxed{\mathcal{F}_0} = \{\phi, \Omega\}$

We interpret σ −algebra as record of information. Suppose an outcome(ω) has happened which we don't know, but we are told if each of the event in $\mathcal{F}_1$ is true or false i.e., whether ω belongs to each elements of $\mathcal{F}_1$ or not. This will lead to the precise knowledge of the first toss. In other words, $\mathcal{F}_1$ stores the information till the first toss.

For example, if the actual outcome(ω) is HHT which we do not know. But we will say $\omega \notin \phi$, $\omega \in \Omega$, $\omega \in A_H$, $\omega \notin A_T$. This information tells us that the first toss was H, nothing more.

σ −algebra

- Similarly, if we define

Ω

$$A_{HH} = \{HHH, HHT\} \text{ ... HH in the first two tosses}$$
$$A_{HT} = \{HTH, HTT\} \text{ ... HT in the first two tosses}$$
$$A_{TH} = \{THH, THT\} \text{ ... TH in the first two tosses}$$
$$A_{TT} = \{TTH, TTT\} \text{ ... TT in the first two tosses}$$

4

$$\mathcal{F}_2 = \{\phi, \Omega, A_{HH}, \boxed{A_{HT}, A_{TH}}, A_{TT}, \\ A_H, A_T, A_{HH} \bigcup A_{TH}, A_{HH} \bigcup A_{TT}, A_{HT} \bigcup A_{TH}, A_{HT} \bigcup A_{TT}, \\ A_{HH}{}^c, A_{HT}{}^c, A_{TH}{}^c, A_{TT}{}^c\}$$

We can see that $\mathcal{F}_2$ records the information till the first two coin tosses.

Filtration

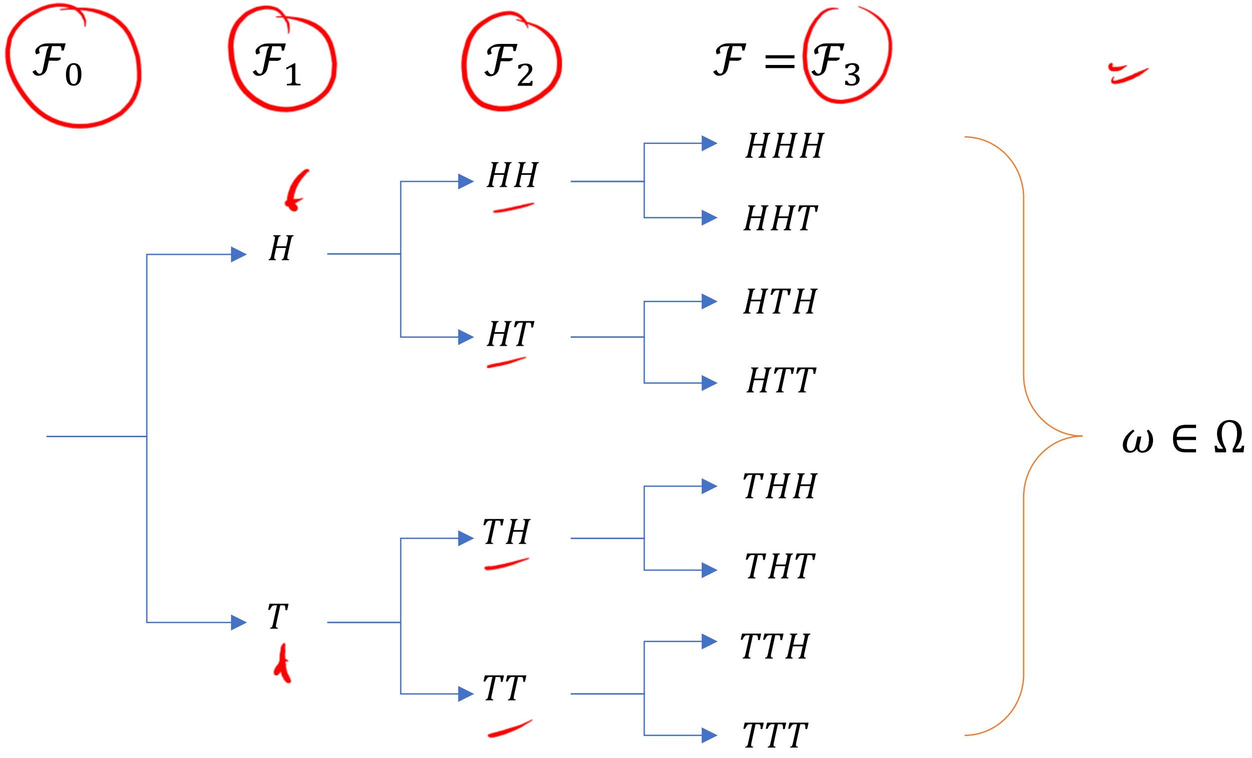
PEAKS2TAILS

- The σ −algebra $\mathcal{F} = \mathcal{F}_3$ which is the collection of all subsets of Ω (Power Set of Ω) contains full information.
- A filtration is a sequence of σ −algebras $\mathcal{F}_0, \mathcal{F}_1, \ldots, \mathcal{F}_n$ such that $\mathcal{F}_i \supset \mathcal{F}_{i-1}$. So, basically all σ −algebra in the sequence contain all the elements of all previous σ −algebras.

$\mathcal{F}_2$

$\{\phi, \Omega, \{HHH\}, \{HHT\}, \ldots \{T,T,T\} \ldots \}$

HHT



Probability Measure

- We understand probability as the chances of an event happening. Formally speaking, a probability measure $\mathbb{P}$ defined on $(\Omega, \mathcal{F})$ is a function that maps $\mathcal{F}$ to $[0,1]$ and has the following properties

Measurable space

$\mathbb{R}$

I. $\mathbb{P}(\phi) = 0$

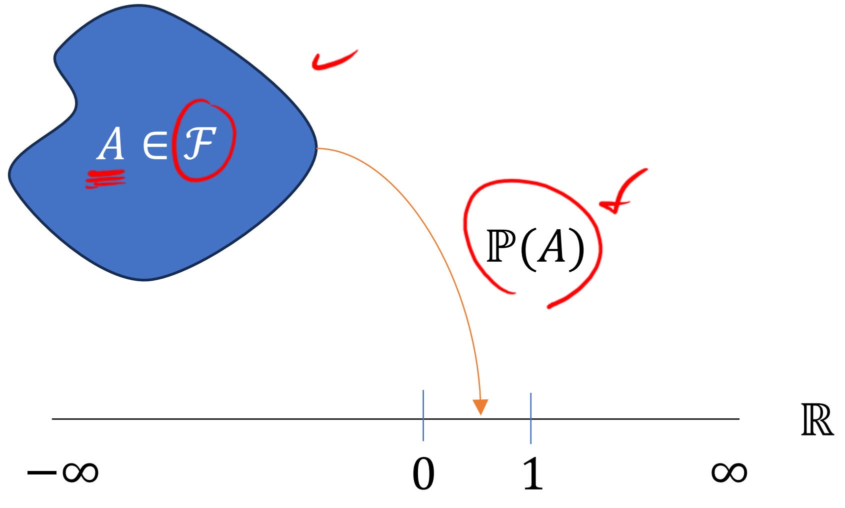
II. $\mathbb{P}(A) \geq 0$ for every $A \in \mathcal{F}$

III. $\mathbb{P}(\Omega) = 1$

IV. if $A_1, A_2, \ldots,$ is a sequence of disjoint sets in $\mathcal{F}$, then

$$\mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mathbb{P}(A_k)$$

mutually exclusive



The triplet $(\Omega, \mathcal{F}, \mathbb{P})$ is called a probability-space

Probability Measure

PEAKS2TAILS

- Example – Assume the probability of H and T in one coin toss be 1/3 and 2/3 respectively, then we can derive the probabilities of all complex events.
- First, we find out the probabilities of all the singletons of Ω. Here we assume all tosses are independent

$\mathbb{P}\{THT, HTT\} =$

$$\mathbb{P}\{HHH\} = \left(\frac{1}{3}\right)^3,\ \mathbb{P}\{HHT\} = \left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right),\ \mathbb{P}\{HTH\} = \left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right),\ \mathbb{P}\{HTT\} = \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^2$$

$\rightarrow$ first toss is head.

$$\mathbb{P}\{THH\} = \left(\frac{2}{3}\right)\left(\frac{1}{3}\right)^2,\ \mathbb{P}\{THT\} = \left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right),\ \mathbb{P}\{TTH\} = \left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right),\ \mathbb{P}\{TTT\} = \left(\frac{2}{3}\right)^3$$

Note that the probability of H in the first toss in an experiment with 3 tosses can be recovered as follows. We use the property IV to add the probabilities of disjoint sets

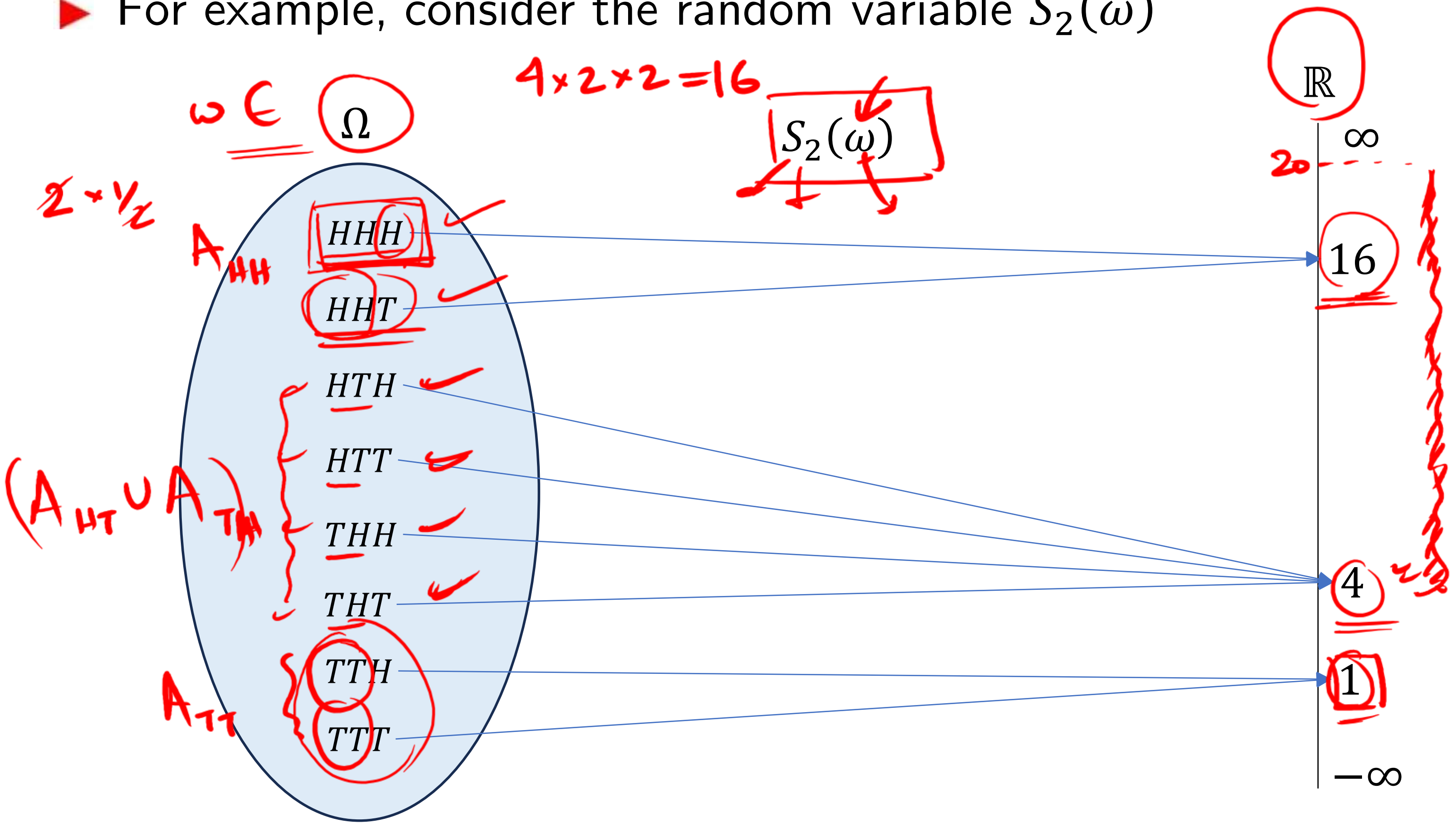
$$\mathbb{P}(A_H) = \mathbb{P}\{HHH, HHT, HTH, HTT\} = \mathbb{P}(\{HHH\} \cup \{HHT\} \cup \{HTH\} \cup \{HTT\})$$
$$= \left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^2 = \frac{1}{3}$$

Random Variable

- Let's consider a wealth process (S). The initial wealth is $S_0 = \$4$. Now with every coin toss the wealth doubles if the outcome is H and halves if the outcome is T. In this process, S_1, S_2 and S_3 are random variables. A random variable is function that is a mapping from $\Omega \longrightarrow \mathbb{R}$

$\omega \in \Omega$

- For example, consider the random variable $S_2(\omega)$



What is the pre-image under S_2 of this interval [4,20] ?

A_{TT}^c

$S_2^{-1}(4,20) = ?$ A_{HH}

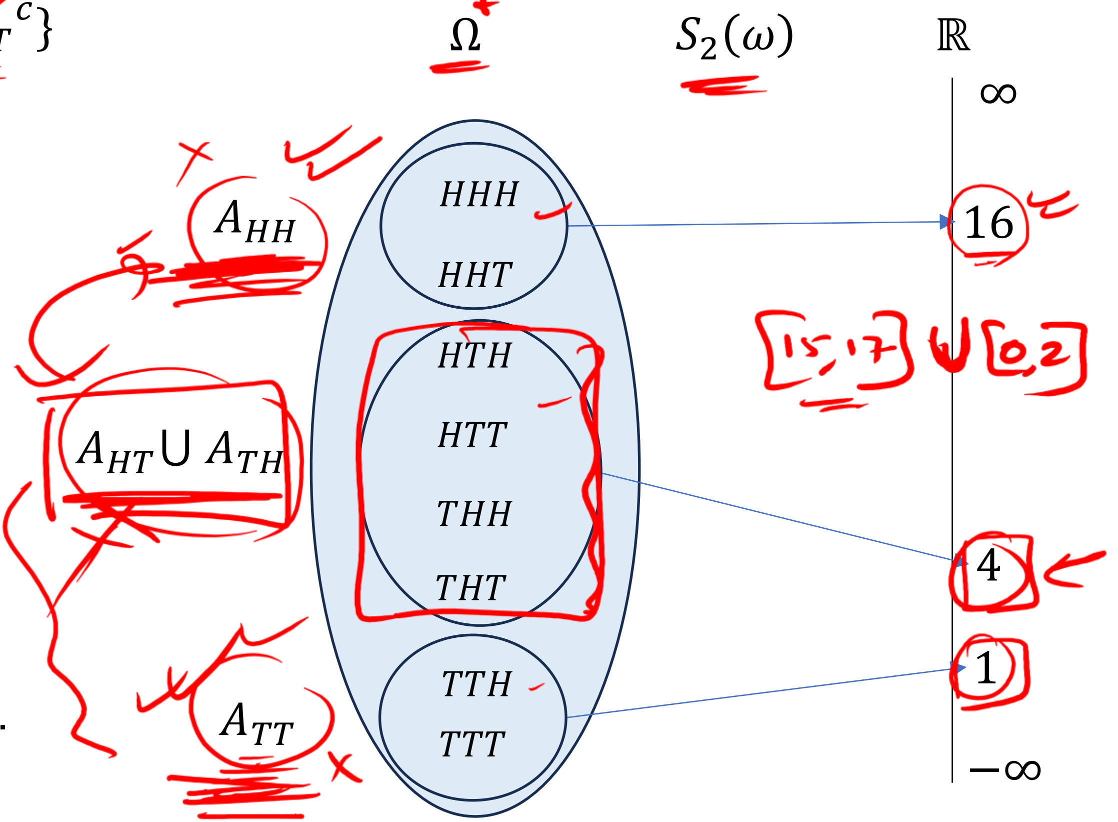
$S_2^{-1}(4,16) = \phi$

σ −algebra generated by a Random Variable

- The complete collection of pre-images under S_2 is given as $\phi, \Omega, A_{HH}, A_{TT}, A_{HT} \cup A_{TH}$ and all possible unions of these. This collection of sets is a σ −algebra and is called the σ −algebra generated by random variable S_2 and is denoted by $\sigma(S_2)$

$[0,2]$ $(\Omega, \mathcal{F})$ $\sigma(X)$

$\mathcal{F}_2 \neq$ $\sigma(S_2) = \{\phi, \Omega, A_{HH}, A_{TT}, A_{HT} \bigcup A_{TH}, A_{HH} \bigcup A_{TT}, {A_{HH}}^c, {A_{TT}}^c\}$



- The above σ −algebra contains the same information that we will get by observing S_2. For example, if we are told that the outcome is not in A_{HH} or A_{TT} but in $A_{HT} \cup A_{TH}$ then we would know that the first two tosses contain one H and one T, nothing more. The same information is contained in saying $S_2 = 4$.

Y_2 = (Sum of heads in the two tosses)

- Note that $\sigma(S_2) \neq \mathcal{F}_2$ as $\mathcal{F}_2$ contains more information. In $\mathcal{F}_2$, we have precise information of the first two tosses.

Measurable

- Let $\mathcal{F}$ be the σ −algebra of all subsets of Ω. Let X be a random variable $\Omega \longrightarrow \mathbb{R}$ and $\sigma(X)$ be the σ −algebra generated by X. We say X is $\mathcal{G}$ − measurable if $\sigma(X) \subseteq \mathcal{G}$

  For Example, we can say S_2 is $\mathcal{F}_2$ measurable but not $\mathcal{F}_1$ measurable

  $\sigma(S_2) \subseteq \mathcal{F}_2$

This simply means that if we know $\mathcal{F}_2$, we precisely know the value S_2

But if we know only $\mathcal{F}_1$, then S_2 is still random

Induced Measure

- Consider the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable $X: \Omega \longrightarrow \mathbb{R}$. Given a set $A \subseteq \mathbb{R}$, we define the induced measure of A to be $\mathcal{L}_X(A) = \mathbb{P}(X \in A)$

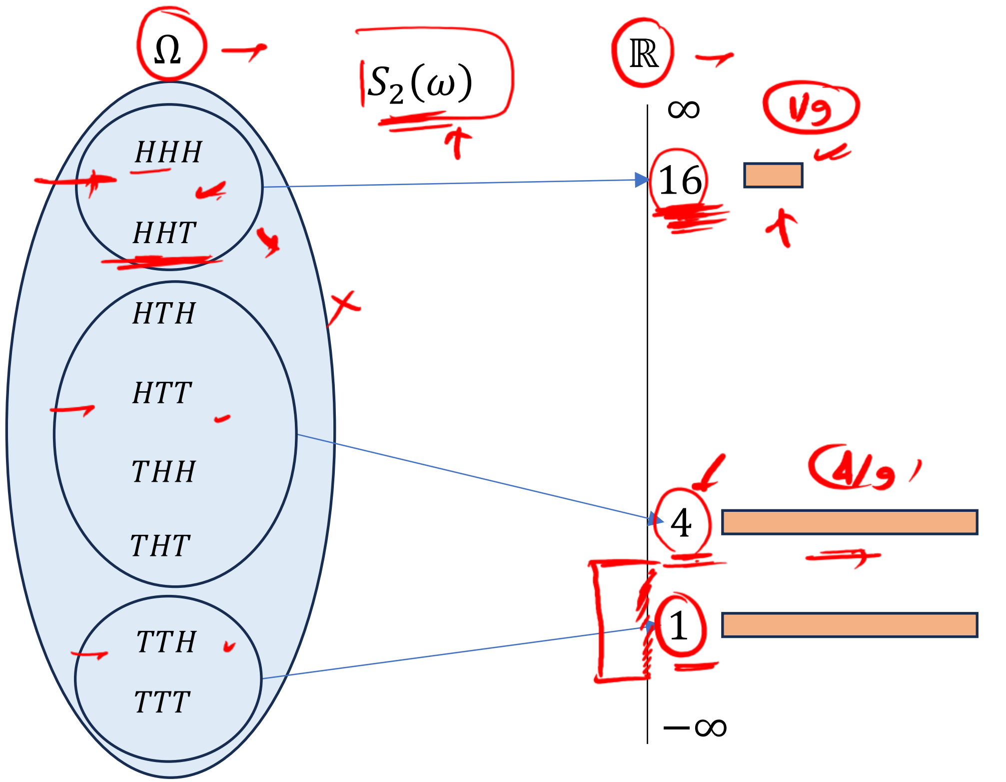
$A = [1,2]$

In other words, it tells us the probability that X takes value in A. For example, consider our random variable S_2. We can place probability masses on $\mathbb{R}$ based on the induced measure.

$\mathbb{P}(A_{HH}) = \frac{1}{9}$

$\mathbb{P}(A_{HT} \cup A_{TH}) = \frac{4}{9}$

$\mathbb{P}(A_{TT}) = \frac{4}{9}$



$\mathcal{L}_{S_2}(\phi) = \mathbb{P}(\phi) = 0$

$\mathcal{L}_{S_2}(\mathbb{R}) = \mathbb{P}(\Omega) = 1$

$\mathcal{L}_{S_2}[0, \infty) = \mathbb{P}(\Omega) = 1$

$\mathcal{L}_{S_2}[0,3] = \mathbb{P}(S_2 = 1) = \mathbb{P}(A_{TT}) = \frac{4}{9}$

Probability Distribution

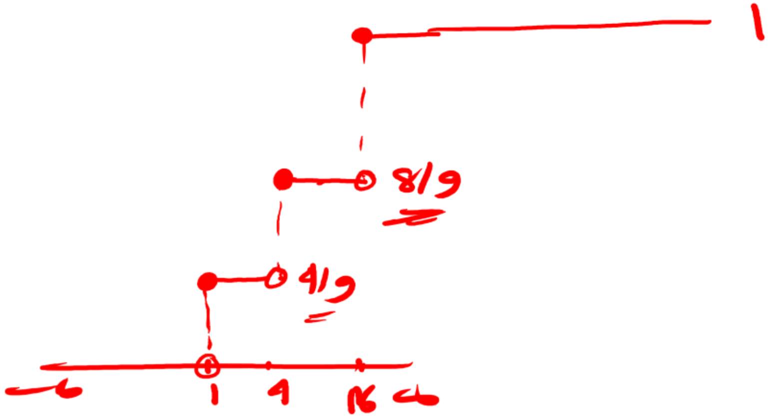
- There are two main ways of recording the information $\boxed{\mathcal{L}_X}$

(Pmf)

$\rightarrow$ Probability mass function $:= f_X(x)$ $\qquad X = (x)$

Cumulative distribution function $:= F_X(x)$

(cdf)



In our example,

$$f_{S_2}(x) = \begin{cases} \frac{4}{9}, & x = 1 \\ \frac{4}{9}, & x = 4 \\ \frac{1}{9}, & x = 16 \end{cases} \qquad F_{S_2}(x) = \begin{cases} 0 & x < 1 \\ \frac{4}{9}, & 1 \le x < 4 \\ \frac{8}{9}, & 4 \le x < 16 \\ 1, & x \ge 16 \end{cases}$$

$F_X(-\infty) = 0$

$F_X(\infty) = 1$

(right-continuous)

$F_X(x^+) = F_X(x)$

Note that random variables and probabilities are separate constructs and should not be mixed.

Expected Value

PEAKS2TAILS

- The expected value of a random variable X is given as

$X(\omega)$

x

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega)\mathbb{P}(\omega)$$

x

The sum is over all outcomes in Ω. Since in our example, X only takes discrete values $(x_1, x_2, \dots, x_k)$ we can rewrite expectation to be a sum over the real number line $\mathbb{R}$

$$\mathbb{E}[X] = \sum_{k=1}^{n} x_k \mathcal{L}_X\{x_k\}$$

pmf

In our example, $\mathbb{E}[S_2] = 1 * \mathcal{L}_{S_2}\{1\} + 4 * \mathcal{L}_{S_2}\{4\} + 16 * \mathcal{L}_{S_2}\{16\} = 1 * \frac{4}{9} + 4 * \frac{4}{9} + 16 * \frac{1}{9} = 4$

Variance

- The variance of a random variable X is given as

$$\mathbb{V}[X] = \sum_{\omega\in\Omega} (X(\omega) - \mathbb{E}[X])^2 \mathbb{P}(\omega)$$

Once again sum is over all outcomes in Ω. We can rewrite expectation to be a sum over the real number line $\mathbb{R}$

$$\mathbb{V}[X] = \sum_{k=1}^{n} (x_k - \mathbb{E}[X])\mathcal{L}_X\{x_k\}$$

In our example, $\mathbb{V}[S_2] = (1 - 4)^2 * \mathcal{L}_{S_2}\{1\} + (4 - 4)^2 * \mathcal{L}_{S_2}\{4\} + (16 - 4)^2 * \mathcal{L}_{S_2}\{16\}$

$$= 9 * \frac{4}{9} + 0 * \frac{4}{9} + 144 * \frac{1}{9} = 4 + 0 + 16 = 20$$

Borel σ −algebra $\mathcal{B}(\mathbb{R})$

- The Borel σ −algebra is the smallest σ −algebra containing all open intervals in $\mathbb{R}$. It is denoted as $\mathcal{B}(\mathbb{R})$. The sets in $\mathcal{B}(\mathbb{R})$ are called Borel sets.

$\mathbb{R} \to \Omega$

$\to \mathcal{F}$

$\Omega = \mathbb{R}$

- Every open interval (a, b) is in $\mathcal{B}(\mathbb{R})$
- Every union of open intervals is in $\mathcal{B}(\mathbb{R})$ … property of σ −algebra
- All half-open lines are in $\mathcal{B}(\mathbb{R})$, e.g., $(a, \infty) = \cup_{n=1}^{\infty}(a, a+n)$, $(-\infty, a) = \cup_{n=1}^{\infty}(a-n, a)$
- The union $(-\infty, a) \cup (b, \infty)$ is also in $\mathcal{B}(\mathbb{R})$
- This means that every closed interval is also in $\mathcal{B}(\mathbb{R})$, $[a, b] = \big((-\infty, a) \cup (b, \infty)\big)^c$

It can be shown that closed-half lines, half-closed intervals, half-open intervals are also in Borel. Every set containing countably finite or infinite numbers are also in Borel.

$\mathbb{R}$ is in

Pretty much any imaginable set in $\mathbb{R}$ is in $\mathcal{B}(\mathbb{R})$. However, not all sets are Borel. (Cantor set)

Borel Measurable

- Consider a function $f : \mathbb{R} \longrightarrow \mathbb{R}$. We say f is Borel-measurable if whenever $B \in \mathcal{B}(\mathbb{R})$, $f^{-1}(B) \in \mathcal{B}(\mathbb{R})$. Indirectly, we are saying that the σ −algebra generated by f should be in $\mathcal{B}(\mathbb{R})$ for the function to be Borel-measurable.

In practice, it's hard to produce functions which are not Borel-measurable. We will assume all sub-sets of $\mathbb{R}$ are Borel sets and all $f : \mathbb{R} \longrightarrow \mathbb{R}$ are Borel-measurable functions.

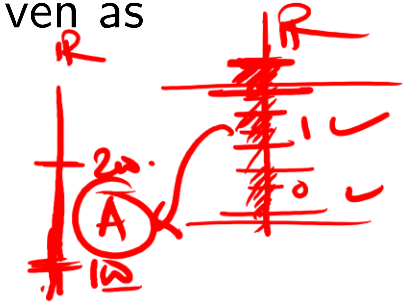
$\mathcal{G}$-measurable

$X \rightarrow \sigma(X) \subset \mathcal{G}$

Borel Measurable

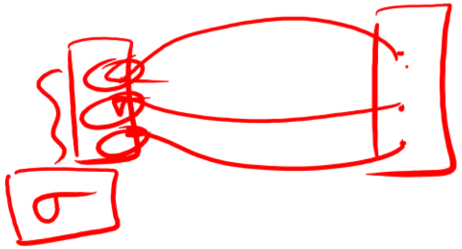
- Example – Consider the indicator function $\mathbb{I}_A(x)$ where $A \subseteq \mathbb{R}$ given as

$$\mathbb{I}_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$



$\mathbb{R} - A = A^c$

Now if we take any $B \in \mathcal{B}(\mathbb{R})$,



$$\mathbb{I}_A^{-1}(x) = \begin{cases} \phi, & 0 \notin B, 1 \notin B \\ A, & 0 \notin B, 1 \in B \\ A^c, & 0 \in B, 1 \notin B \\ \mathbb{R}, & 0 \in B, 1 \in B \end{cases}$$

We can see $\{\phi, A, A^c, \mathbb{R}\}$ is a σ −algebra contained in $\mathcal{B}(\mathbb{R})$, so $\mathbb{I}_A(x)$ is Borel-measurable.

Borel Measurable

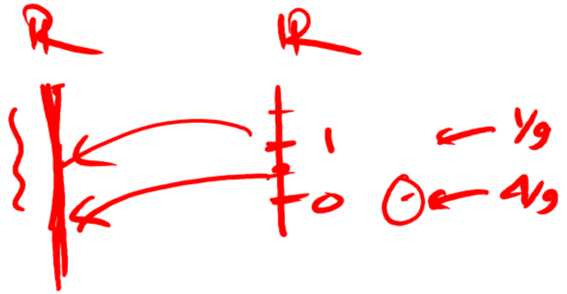
Exercise - 1. What is $\mathbb{E}[\mathbb{I}_A(x)]$ **?**

Exercise - 2. Examine the Borel-measurability for the indicator function $\mathbb{I}_{x\geq 100}$

Exercise – 3. Are there any functions which are measurable on $(\phi, \mathbb{R})$?

$$\mathbb{E}[\mathbb{I}_A(x)] = ? \qquad 0 \times \mathbb{P}(A^c) + 1 \times \mathbb{P}(A) = \mathbb{P}(A)$$

$$\mathbb{I}_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$



$$\sigma(f) = \{\phi, \mathbb{R}\}$$

measurable space

$$f: \mathbb{R} \to \mathbb{R}$$

$$f(x) = c \qquad \text{Const}$$

$$5$$

Measure

- A Measure defined on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is a function $\mu: \mathcal{B} \longrightarrow [0, \infty]$ with the following properties

I. $\mu(\phi) = 0$

II. if $A_1, A_2, \ldots,$ is a sequence of disjoint sets in $\mathcal{F}$, then $\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mu(A_k)$

- A Lebesgue measure is defined to be a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ which assigns the measure of each interval to be its length. Let's denote it as μ_0
- A measure has all the properties of Probability except that total measure of the space is not 1. Note that $\mu_0(\mathbb{R}) = \infty$
- The Lebesgue measure of a set containing only one point is 0 , i.e., $\mu_0\{a\} = 0$
- The Lebesgue measure of a countably infinite set is also 0 , i.e., $\mu_0\{a_1, a_2, \ldots\} = 0$
- For uncountably infinite number of points, Lebesgue measure is can be 0, positive finite or infinite. We cannot simply add the measure for all uncountable points. Integration comes to the rescue !!